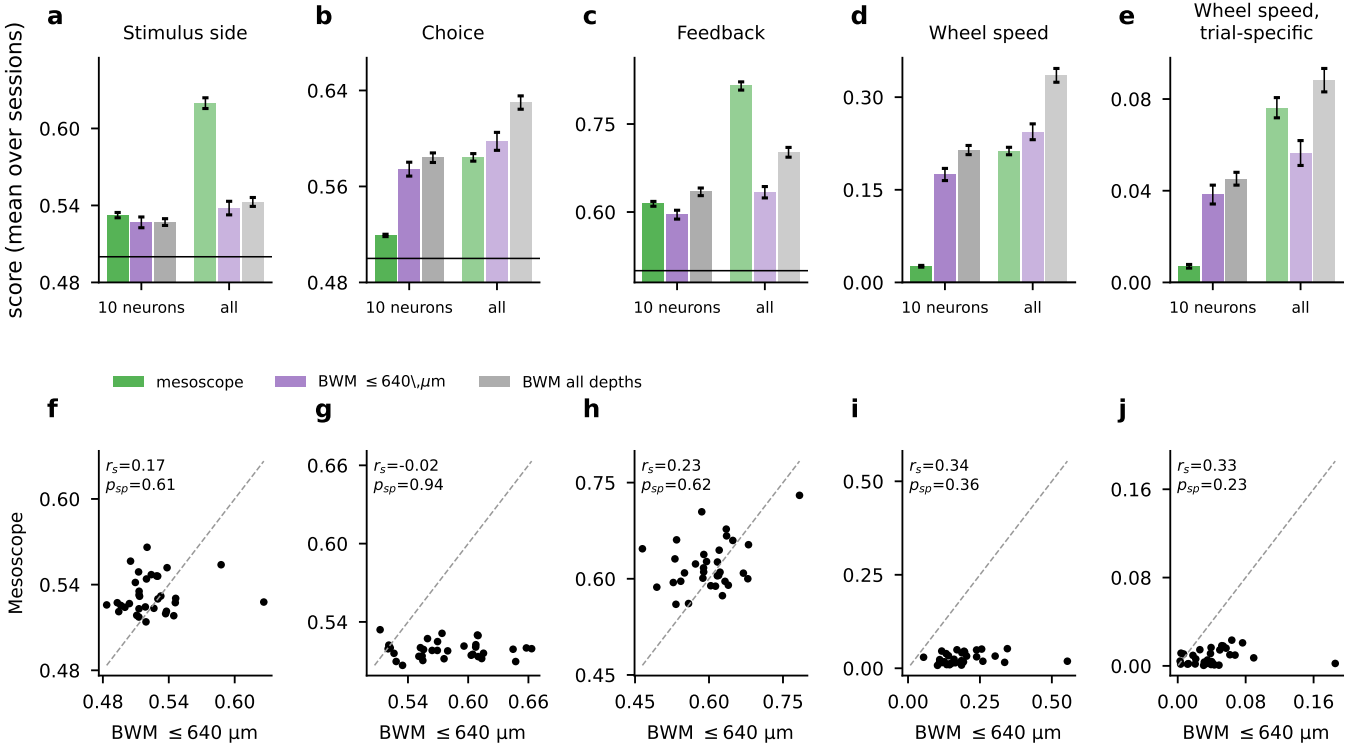

Figure 1: Summary of the revised decoding. **a–e** Mean over sessions (± s.e.m.) of each session's mean score over its 1 mm bins, from 10 random neurons (left, solid) and all neurons (right, light). **f–j** Per 1 mm bin, 10 neurons: BWM $\leq$640 μm (x) vs mesoscope (y), the 33 bins covered by both; Spearman r_s, naive p, and p_{sp} against 1000 variogram-matched surrogates of either map (larger p reported). Decoding per session and bin (sessions with $\geq$50 trials and $\geq$10 neurons in the bin), 5-fold cross-validation over trials. Binary variables: L2 logistic regression, balanced accuracy (chance 0.5). Wheel speed in six 0.2 s bins from -0.2 to 1 s around movement onset: ridge regression on the activity of the previous, same and next bin (penalty by exact leave-one-trial-out error), cross-validated R^2; trial-specific: the same with the mean speed profile of the training trials subtracted, so only trial-to-trial differences in speed count. **a** Stimulus side, 10 neurons: mesoscope 0.532, BWM $\leq$640 μm 0.527, BWM all depths 0.527; all neurons: mesoscope 0.620, BWM $\leq$640 μm 0.538, BWM all depths 0.543. **b** Choice, 10 neurons: mesoscope 0.519, BWM $\leq$640 μm 0.574, BWM all depths 0.584; all neurons: mesoscope 0.584, BWM $\leq$640 μm 0.598, BWM all depths 0.630. **c** Feedback, 10 neurons: mesoscope 0.614, BWM $\leq$640 μm 0.595, BWM all depths 0.634; all neurons: mesoscope 0.815, BWM $\leq$640 μm 0.634, BWM all depths 0.702. **d** Wheel speed, 10 neurons: mesoscope 0.026, BWM $\leq$640 μm 0.175, BWM all depths 0.214; all neurons: mesoscope 0.213, BWM $\leq$640 μm 0.244, BWM all depths 0.335. **e** Wheel speed, trial-specific, 10 neurons: mesoscope 0.007, BWM $\leq$640 μm 0.038, BWM all depths 0.045; all neurons: mesoscope 0.076, BWM $\leq$640 μm 0.056, BWM all depths 0.088. **f** Stimulus side: r_s = 0.17, naive p = 0.34, p_{sp} = 0.61. **g** Choice: r_s = -0.02, naive p = 0.92, p_{sp} = 0.94. **h** Feedback: r_s = 0.23, naive p = 0.21, p_{sp} = 0.62. **i** Wheel speed: r_s = 0.34, naive p = 0.05, p_{sp} = 0.36. **j** Wheel speed, trial-specific: r_s = 0.33, naive p = 0.06, p_{sp} = 0.23.